REG.NO.:

VIT
Vellore Institute of Technology

School of Computer Science and Engineering

CONTINUOUS ASSESSMENT TEST - II
FALL SEMESTER 2024-2025

SLOT:A1

Programme Name & Branch :B.Tech -CSE
Course Code and Course Name : BCSE305L-Embedded Systems
Faculty Name(s) : Dr.Bhulakshmi B **Class Number(s):** VL2024250108125
Date of Examination :13-10-2024
Exam Duration : 90 minutes **Maximum Marks: 50**

Answer ALL the questions

Q. No	Question	M
1.	How do the design requirements and constraints differ between hard and soft real time systems? **Provide** an example for both type of systems.	10
2.	**Evaluate** how the following parameters affect the schedulability of real time tasks:- • Arrival Time • Current Time / Scheduling point • Execution Time • Rate or Period • Deadline Suggest an **optimal scheduling** scheme using any three parameters mentioned above and apply it for the following dataset:-	10

Task	Arrival Time	Execution Time	Period	Deadline
T1	0	4	20	8
T2	0	3	5	5
T3	0	3	10	9

Note: Consider current time as per the scheduling points. Illustrate the task time-line graph for two cycles.

3.	Find out the needs for software optimisation techniques and analyse various Software optimisation techniques in details with respect to embedded system software design using suitable examples.			10
4.	**Compute** the CPU utilization and check whether the given tasks are schedulable in RTOS. Apply **RMS** and **EDF** algorithms on the given data. **Illustrate** the task time-line graph for two cycles.			10

Task	Start time	Execution Time	Deadline	Period
T1	0	10	30	30
T2	4	3	10	10
T3	5	10	25	25

-PTO-

5.	Construct **FSM model And CDFG** for the given real time embedded system. **"Industrial Temperature Control System"** used in industrial environments to maintain temperature within safe operational limits and avoid equipment damage, product spoilage, or safety hazards. **Requirements:** • The system must accurately measure temperature. • Temperature readings must be updated and displayed in real-time. • Threshold Alarms- an audible and/or visual alarm must be triggered when the temperature exceeds these thresholds. • Relevant states, events and actions to be considered.	10

****ALL THE BEST****